

Question Paper

Exam Date & Time: 28-Dec-2025 (09:30 AM - 12:30 PM)



MANIPAL ACADEMY OF HIGHER EDUCATION

MANIPAL INSTITUTE OF TECHNOLOGY - BENGALURU
EIGHTH SEMESTER B.TECH. DEGREE MAKEUP EXAMINATIONS - DECEMBER 2025

Advanced Computer Networks [CSE_5151H]

Marks: 50

Duration: 180 mins.

PART-A

Answer all the questions.

- 1) Drones in a high-speed delivery network frequently change positions and altitudes. Analyse the suitability of AODV routing over DSR for maintaining stable connections. (5)
- A) Course Outcome: "CO1" / Cognitive Level: "Analysis"
- B) A forest monitoring system uses multiple drones to detect wildfires. Apply multi-UAV coordination to improve coverage and reliability. (5)
- Course Outcome: "CO1" / Cognitive Level: "Application"
- 2) A smart manufacturing firm integrates numerous IoT devices that demand dynamic bandwidth allocation. Examine how Software-Defined Networking (SDN) employs policy-driven control to enhance network responsiveness and optimize IoT device management (5)
- A) Course Outcome: "CO2" / Cognitive Level: "Understand"
- B) A telecom operator deploys new services requiring multi-vendor compatibility. Apply open APIs and protocols like OpFlex to achieve interoperability and flexibility in SDN. (5)
- Course Outcome: "CO2" / Cognitive Level: "Application"
- 3) A hospital's data centre faces inefficiency due to each application using a dedicated server. Analyse how moving from direct attached to shared storage over SAN improves resource use. (5)
- A) Course Outcome: "CO3" / Cognitive Level: "Analysis"
- B) A company wants to integrate compute, storage, and networking into a single platform. Apply the concept of hyperconvergence to simplify infrastructure management. (5)
- Course Outcome: "CO3" / Cognitive Level: "Application"
- 4) A content provider must serve videos efficiently to users across continents. Apply suitable CDN deployment models such as "Enter Deep" or "Bring Home" to optimize global content delivery performance. (5)
- A) Course Outcome: "CO4" / Cognitive Level: "Application"
- B) Users complain of distorted audio during peak traffic. (5)
- Apply jitter buffering and adaptive playout strategies to stabilize voice quality.

Course Outcome: "CO4" / Cognitive Level: "Application"

- 5) A network must guarantee bandwidth for VIP video calls while rejecting excess load. Analyse how call admission control preserves QoS for existing users under limited capacity. (5)

A)

Course Outcome: "CO4" / Cognitive Level: "Analysis"

- B) A telecom company expands its fiber backbone to support multiple high-speed services. Explain the WDM architecture that enables efficient bandwidth utilization and scalability in optical networks. (5)

Course Outcome: "CO5" / Cognitive Level: "Understand"

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Event Name	Subject
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058-MIT BLR-MAKEUP EXAM-2025-2026

CSE_5151H:Advanced Computer Networks

#	Course Outcome	Description	Marks
1	1	CO 1	10
2	2	CO 2	10
3	3	CO 3	10
4	4	CO 4	15
5	5	CO 5	5
Total			50

* **NA** - CO Not Selected

Event Name	Subject
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058-MIT BLR-MAKEUP EXAM-2025-2026

CSE_5151H:Advanced Computer Networks

#	Level	No of Questions	Marks (%)
1	Remember	0	0(0%)
2	Understand	2	10(20%)
3	Application	5	25(50%)
4	Analysis	3	15(30%)
5	Create	0	0(0%)
6	Evaluate	0	0(0%)
Total		10	50 (100%)